Artificial Intelligence & RAG Knowledge Base

Structured test document for the AI Document Chatbot

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science focused on developing systems that can perform tasks associated with human intelligence, including learning, reasoning, perception, language understanding, planning, and decision making. AI is used in image recognition, recommendation systems, healthcare, fraud detection, robotics, and intelligent assistants.

2. Machine Learning

Machine Learning (ML) is a major area of AI in which algorithms learn patterns from data rather than being explicitly programmed for every decision. Supervised learning uses labeled examples, unsupervised learning discovers patterns in unlabeled data, and reinforcement learning learns through interaction and rewards.

3. Deep Learning

Deep Learning is a subfield of machine learning that uses neural networks with multiple layers. ANN models are useful for structured data, CNNs learn spatial patterns, while RNNs and LSTMs process sequential information. Deep Learning is widely used in computer vision, speech processing, and language applications.

4. Natural Language Processing

Natural Language Processing (NLP) enables computers to process, understand, analyze, and generate human language. Applications include sentiment analysis, text classification, translation, summarization, information extraction, question answering, and conversational assistants.

5. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. A RAG system retrieves relevant information from external documents and supplies it to a language model as context. This is useful for private documents, research collections, organizational knowledge bases, and changing information.

6. Large Language Models

Large Language Models (LLMs) are neural language models trained on large collections of text. They can understand questions, summarize information, generate explanations, and perform language tasks. In a document chatbot, the LLM receives a question together with relevant retrieved context and generates the response.

7. Vector Embeddings

Embeddings are numerical representations of text that capture semantic meaning. Documents are divided into chunks and each chunk is converted into a vector. Similar meanings tend to have similar representations, allowing semantic search to retrieve relevant content.

8. Vector Databases

A vector database stores embeddings and supports similarity search. In a RAG application, document chunks and their embeddings are stored in the database. A user's question is embedded and compared with stored vectors so that the most relevant chunks can be retrieved.

9. ChromaDB

ChromaDB is a vector database commonly used in AI and RAG applications. It stores document embeddings together with metadata such as source file names. A chatbot can use ChromaDB to index uploaded documents and retrieve semantically similar chunks during question answering.

10. LangChain

LangChain is a framework for building applications around language models. It provides components for document loading, text splitting, embeddings, vector stores, retrievers, prompts, and conversational workflows. It can coordinate the movement of information from uploaded documents through retrieval and finally to the LLM.

11. Document Processing Pipeline

A document-based AI assistant normally loads files, extracts content, splits large texts into manageable chunks, creates embeddings, stores vectors in a vector database, retrieves relevant chunks for each question, places them into a prompt, and sends the prompt to an LLM.

12. Prompt Engineering

Prompt engineering is the design of instructions given to a language model. A document chatbot should instruct the model to answer using retrieved context and to state when requested information is not available. Good prompts improve relevance, consistency, and control over generated responses.

13. Conversational Question Answering

A conversational RAG chatbot should understand follow-up questions. After asking about Artificial Intelligence, a user might ask, 'What are its applications?' Conversation history helps resolve references such as 'it' or 'this approach'.

14. Applications of AI

Artificial Intelligence is used in healthcare, education, finance, cybersecurity, manufacturing, transportation, agriculture, customer support, and scientific research. Examples include medical image analysis, personalized learning, fraud detection, predictive maintenance, recommendation engines, and intelligent document assistants.

15. Challenges and Limitations

AI systems can face biased data, incomplete information, hallucinated responses, privacy concerns, computational cost, and limited interpretability. RAG can reduce some factual errors by grounding responses in retrieved documents, but retrieval quality and document quality remain important.

16. Summary

Artificial Intelligence combines techniques for intelligent information processing. Machine Learning enables systems to learn from data, Deep Learning uses multi-layer neural networks, NLP focuses on human language, and RAG connects retrieval systems with LLMs. Embeddings, vector databases, LangChain, and LLMs can work together to build practical document question-answering systems.